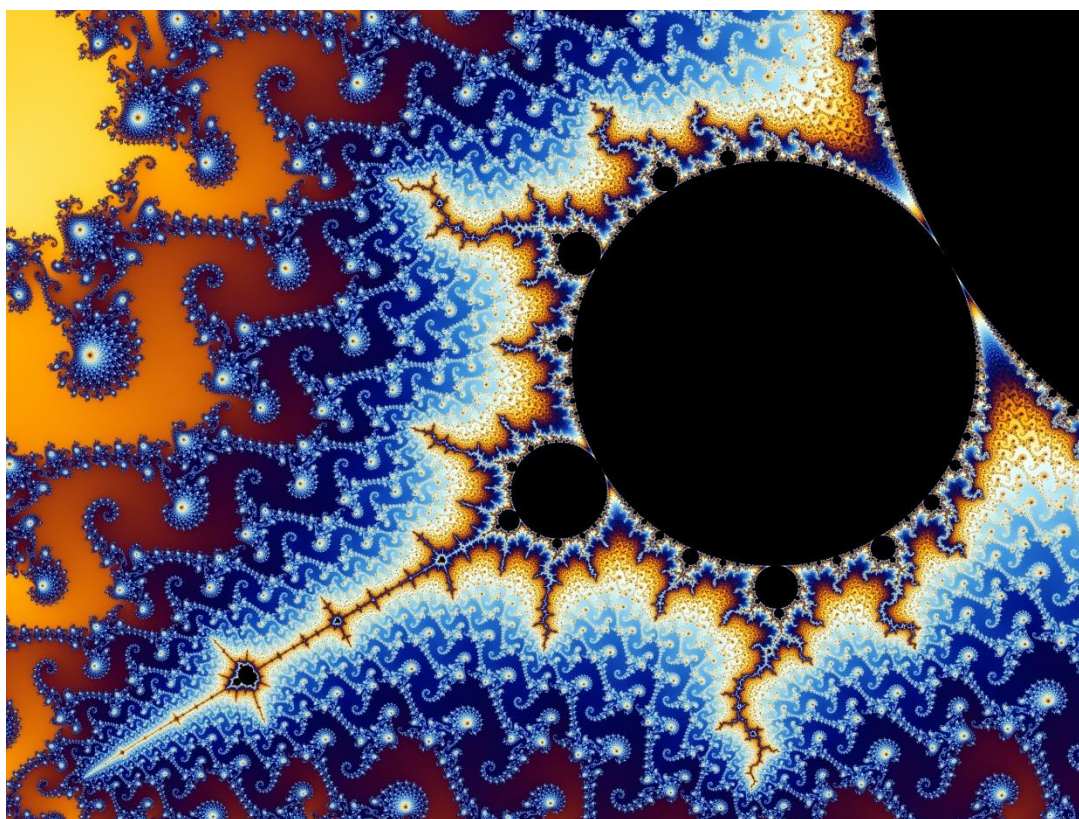




Tour de Maths 2025

Leg 1

Question Paper - MEMO



The Mandelbrot set

5 Mark Questions

Question 1 In the following equation, $\sqrt{CASIO} = 304$, each letter represents a unique digit from the digits 0-9. What is the digit value of the letter S?	MEMO $\sqrt{CASIO} = 304$ $CASIO = 92416$ $S = 4$ ANSWER: 4
Question 2 How many decimals are there after the decimal point when we calculate $\frac{2025}{2^{16}}$? Note: your calculator truncates this answer so don't trust it entirely	MEMO Because the denominator is power of 2 it will always terminate. In this case it terminates on the number 0,0308990478515625 which is 16 digits long, the same as the power of the denominator ANSWER: 16
Question 3 What is the last digit if we expand 3^{2025} ?	MEMO Observing the powers of 3: $3^1 = 3$ $3^2 = 9$ $3^3 = 27$ $3^4 = 81$ The last digits follow a repeating cycle every 4 terms: 3, 9, 7, 1. Since the cycle repeats every 4 numbers, we compute: $2025 \div 4 = 506 \text{ rem } 1$ Since the remainder is 1, the last digit follows the first position in the cycle, which is 3. ANSWER: 3

Question 4 $2025 = 1^n + 2^n + 3^n + \dots + 8^n + 9^n$ Determine n.	MEMO The sum of consecutive cubes is equal to the square of the sum. $2025 = 1^n + 2^n + 3^n + \dots + 8^n + 9^n$ $45^2 = 1^n + 2^n + 3^n + \dots + 8^n + 9^n$ $(1 + 2 + 3 + \dots + 8 + 9)^2 = 1^3 + 2^3 + 3^3 + \dots + 8^3 + 9^3$ ANSWER: 3
Question 5 How many perfect square factors does 2025 have?	MEMO $5^2 \cdot 3^4 = 2025$ $5^2 = 25$ $3^2 = 9$ $5^2 \cdot 3^2 = 225$ These are all the perfect square factors ANSWER: 4
Question 6 What is the value of x in $\sqrt[3]{2025 \times x}$ that will result in an output with the smallest positive integer value?	MEMO $\sqrt[3]{2025 \times x} = \sqrt[3]{5^2 \cdot 3^4 \times x}$ Need powers to be multiples of three. So need a $5 \cdot 3^2 = 45$ ANSWER: 45
Question 7 The sum of the digits of 2025 is a perfect square. Which is the next calendar year that has this property?	MEMO The next year that fulfils this requirement is 2034 $2 + 0 + 3 + 4 = 9$ ANSWER: 2034

Question 8 Determine the value of this power tower, where each digit is the exponent of the digit before itself. $2^{4^{20^{25}}} + 6^{2^{20^{25}}} + 3^{3^{20^{25}}} = ?$	MEMO $2^{4^{20^{25}}} + 6^{2^{20^{25}}} + 3^{3^{20^{25}}}$ $= 2^4 + 6^1 + 3^3$ $= 16 + 6 + 27$ $= 49$ ANSWER: 49
---	--

Question 9 $\frac{2025}{(1+2+3+..+n)^p} = 1$ Determine $n + p$	MEMO The sum of consecutive cubes is equal to the square of the sum. $\frac{2025}{(1+2+3+..+n)^p} = 1$ $2025 = (1+2+3+..+n)^p$ $45^2 = (1+2+3+..+n)^p$ $(1+2+3+..+9)^2 = (1+2+3+..+n)^p$ $n + p = 11$ ANSWER: 11
--	---

Question 10 Would you rather have 20% of a pizza slice with radius 25cm, or would you choose 25% of a pizza slice with radius 20cm? Explain your answer.	MEMO We need to compare two portions of pizza and determine which is larger: 20% of a pizza with radius 25 cm 25% of a pizza with radius 20 cm Step 1: Calculate the Area of Each Pizza - Pizza 1 (radius 25 cm): 625π - Pizza 2 (radius 20 cm): 400π Step 2: Compute the Portions 20% of Pizza 1: 125π 25% of Pizza 2: 100π Step 3: Compare the Portions Since: $125\pi > 100\pi$ the 20% of the larger pizza (25 cm radius) gives more pizza than 25% of the smaller pizza (20 cm radius). Conclusion: I would rather have 20% of the pizza with a 25 cm radius because it gives me a larger portion of pizza. 🍕 ANSWER: 20% of 25cm
--	--

10 Mark Questions

Question 11 How many numbers between 1 and 1000 are divisible by 20 and 25?	MEMO Step 1: Find the Least Common Multiple (LCM) $LCM(20, 25) = 2^2, 5^2 = 100$ Step 2: Count Multiples of 100 in the Range [1, 1000] $\frac{1000}{100} = 10$ Conclusion: There are 10 numbers between 1 and 1000 that are divisible by both 20 and 25. ANSWER: 10
--	--

Question 12 Find the sum of the first 170 terms of the sequence: 2,0,2,5,2,0,2,5,2,0,2,5,2,0,2,5,2..	MEMO Step 1: Compute the Sum of One Full Cycle Each cycle consists of four terms: $2 + 0 + 2 + 5 = 9$ Step 2: Count the Number of Complete Cycles Since we need the sum of the first 170 terms, we determine how many full cycles fit in 170: $\frac{170}{4} = 42 \text{ rem } 2$ Step 3: Compute the Sum of 42 Full Cycles $42 \times 9 = 378$ Step 4: Add the Remaining Terms $2 + 0 = 2$ Step 5: Compute the Total Sum $378 + 2 = 380$ ANSWER: 380
---	---

Question 13

Starting at 2025 and counting backwards in sevens, Psryliop counts 2025; 2018; 2011; 2004; ...and so on.

Which of the following numbers will he end up counting: 49 ; 48 ; 47 ; 46 ; or 45?

MEMO**Step 1: Find the General Formula**

The sequence follows the pattern:

$$T_n = 2025 - 7(n-1) = -7n + 2032$$

Step 2: Check for Given Numbers

$$-7n + 2032 = x \quad \text{where we test } x\{49, 48, \dots, 45\}$$

Step 3: Solve for n When $x = 49$

Since n is a whole number, **49 appears in the sequence.**

Since only **49** is the only that results in a whole number, the final answer is:

ANSWER: 49**Question 14**

The integers are arranged in 6 columns as shown.

1	2	3	4	5	6
12	11	10	9	8	7
13	14	15	16	17	18
					19

In which column does 2025 appear?

MEMO**Step 1: Identify the Row Structure**

Each full cycle consists of **12 numbers** (6 forward + 6 backward). This means that the number **2025** appears somewhere in this repeating 12-number cycle.

Step 2: Find the Row Number

We determine which row contains **2025** by dividing it by **12**:

$$2025 \div 12 = 168 \text{ rem } 9$$

This means **2025** is in row **169** (since remainder 9 places it in the next row).

Step 3: Determine the Column

- In **odd-numbered rows** (like row 169), numbers increase left to right.
- The remainder **9** means **2025 is the 9th number in its row.**
- The column layout is **1 to 6**, then cycles back.
- The 9th number corresponds to **column 3.**

ANSWER: 3

Question 15 13 different positive integers add up to 2025. What is the largest possible number that be used in this addition?	MEMO Step 1: Use the Smallest Possible Integers First To maximize the largest number, we should minimize the sum contribution from the other 12 numbers. The smallest 12 distinct positive integers are: 1,2,3,4,5,6,7,8,9,10,11,12 The sum of these numbers is: $S = \frac{12(12+1)}{2} = 78$ Step 2: Find the Largest Possible Number Let the largest number be x. Then, the equation is: $x + 78 = 2025$ $x = 1947$ ANSWER: 1947
--	---

Question 16 Suppose p and q are positive integers with $20 \leq p < q \leq 25$. How many pairs (p,q) will satisfy this inequality?	MEMO Step 1: Identify Possible Values for p and q $p \in \{20, 21, 22, 23, 24, 25\}$ $q \in \{21, 22, 23, 24, 25\} \text{ for } p = 20$ $q \in \{22, 23, 24, 25\} \text{ for } p = 21$ $q \in \{23, 24, 25\} \text{ for } p = 22$ $q \in \{24, 25\} \text{ for } p = 23$ $q \in \{25\} \text{ for } p = 24$ Step 2: Count Valid Pairs (p,q) Total number of valid pairs: $5+4+3+2+1=15$ ANSWER: 15
---	---

Question 17

What is the smaller angle between the hour hand and minute of a clock at 20h25?

MEMO**Step 1: Calculate the Angle of the Minute Hand**

The minute hand moves **360°** in **60 minutes**, so its angular displacement per minute is:

$$\frac{360}{60} = 6^\circ / \text{min}$$

At **25 minutes**, the minute hand is at:

$$25 \times 6 = 150^\circ$$

Step 2: Calculate the Angle of the Hour Hand

The hour hand moves **360°** in **12 hours**, so its angular displacement per hour is:

$$\frac{360}{12} = 30^\circ / \text{hr}$$

At **8 hours**, the hour hand is at:

$$8 \times 30 = 240^\circ$$

However, the hour hand also moves as time progresses. Since it moves **30° per hour**, it moves:

$$\frac{30}{60} \times 25 = 12,5^\circ$$

Thus, at **8:25 PM**, the hour hand is at:

$$240 + 12,5 = 252,5^\circ$$

Step 3: Find the Difference Between the Two Angles

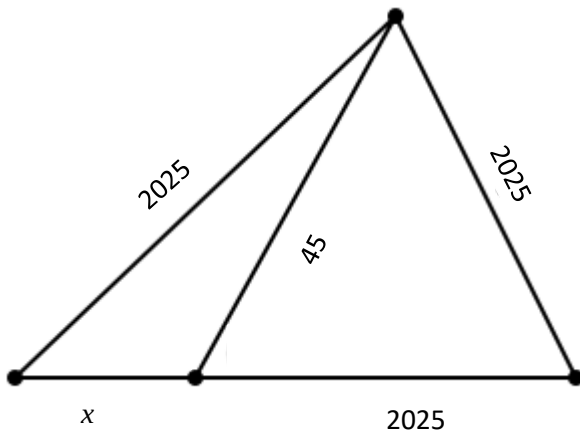
The absolute difference between the hour and minute hand is:

$$|252,5 - 150| = 102,5^\circ$$

ANSWER: 102,5

Question 18

Using the above knowledge, apply it to the scenario below and determine x



MEMO

$$b^2m + c^2n = a(d^2 + mn)$$

$$2025^2 \cdot 2025 + 2025^2 x = (2025 + x)(45^2 + 2025x)$$

$$2025^2(2025 + x) = (2025 + x)(2025 + 2025x)$$

$$2025^2 = 2025 + 2025x$$

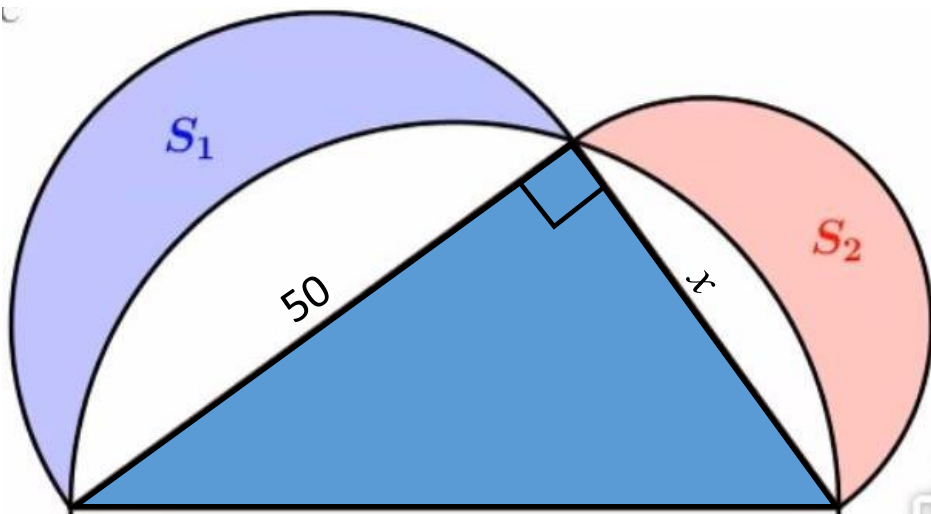
$$2025 = 1 + x$$

$$x = 2024$$

ANSWER: 2024

Question 19

Given the following Geometric property:



If the $S_1 + S_2 = 2025$ determine x .

MEMO

$$S_1 + S_2 = 2025$$

$$\frac{1}{2}(50)x = 2025$$

$$25x = 2025$$

$$x = 81$$

ANSWER: 81

20 Mark Questions

Question 20 Four digit numbers like 2025 and 5220 use the digits 5 once, 2 twice, and 0 once. How many such 4 digit numbers are there? (0225 is not allowed)	MEMO Step 1: Total Arrangements without Restrictions If there were no restrictions on the leading digit, the total number of ways to arrange the digits 5, 2, 2, 0 would be: $\frac{4!}{1!2!1!} = \frac{24}{2} = 12$ This represents the total number of arrangements if there were no restrictions, but some of these might start with 0. Step 2: Subtract the Invalid Cases (Leading Digit is 0) If the number starts with 0, the remaining digits to arrange are 5, 2, 2. The number of ways to arrange these three digits is: $\frac{3!}{2!} = \frac{6}{2} = 3$ Step 3: Valid Cases Now, subtract the invalid cases from the total arrangements: $12 - 3 = 9$ ANSWER: 9
Question 21 2025 is the sum of 5 consecutive integers – what are they?	MEMO Step 1: Represent the Consecutive Integers Let the middle integer be xxx. The five consecutive integers can be written as: $x - 2; x - 1; x; x + 1; x + 2$ Step 2: Sum the Integers The sum of these five integers is: $(x - 2) + (x - 1) + x + (x + 1) + (x + 2) = 2025$ $5x = 2025$ $x = 405$ Step 3: Identify the Five Integers The five consecutive integers are 403, 404, 405, 406, and 407. ANSWER: 403, 404, 405, 406, and 407

Question 22 A school pool has dimensions 20m-by-25m. If we have 5mm of rain, how many litres of water went into the pool?	MEMO Step 1: Convert the Rain Depth to Meters We are given that the rain is 5 mm. To work in consistent units, we convert millimeters to meters: $5mm = 0,005m$ Step 2: Find the Volume of Water Added to the Pool The pool has dimensions 20 meters by 25 meters. The volume of water added to the pool due to 5 mm of rain is the area of the pool multiplied by the depth of the rain: $\begin{aligned} Vol &= lbh \\ &= 20 \times 25 \times 0,005 \\ &= 2,5m^3 \end{aligned}$ Step 3: Convert the Volume to Liters Since 1 cubic meter is equal to 1000 liters, we convert the volume to liters: $2,5m^3 \times 1000 = 2500l$ ANSWER: 2500
--	--

Question 23 $2025 = 3^2 + 4^2 + x^2 + 16^2 + y^2 + 32^2$ Find the value of $x + y$	MEMO $2025 = 3^2 + 4^2 + x^2 + 16^2 + y^2 + 32^2$ $2025 = x^2 + y^2 + 1305$ $x^2 + y^2 = 720$ $x = \pm 24 ; y = \pm 12$ $x = \pm 12 ; y = \pm 24$ $(x + y) = \pm 36$ ANSWER: ± 36
--	---

Total: 200

With special thanks to the sponsorship of

CASIO®